

The Power of Vaccination

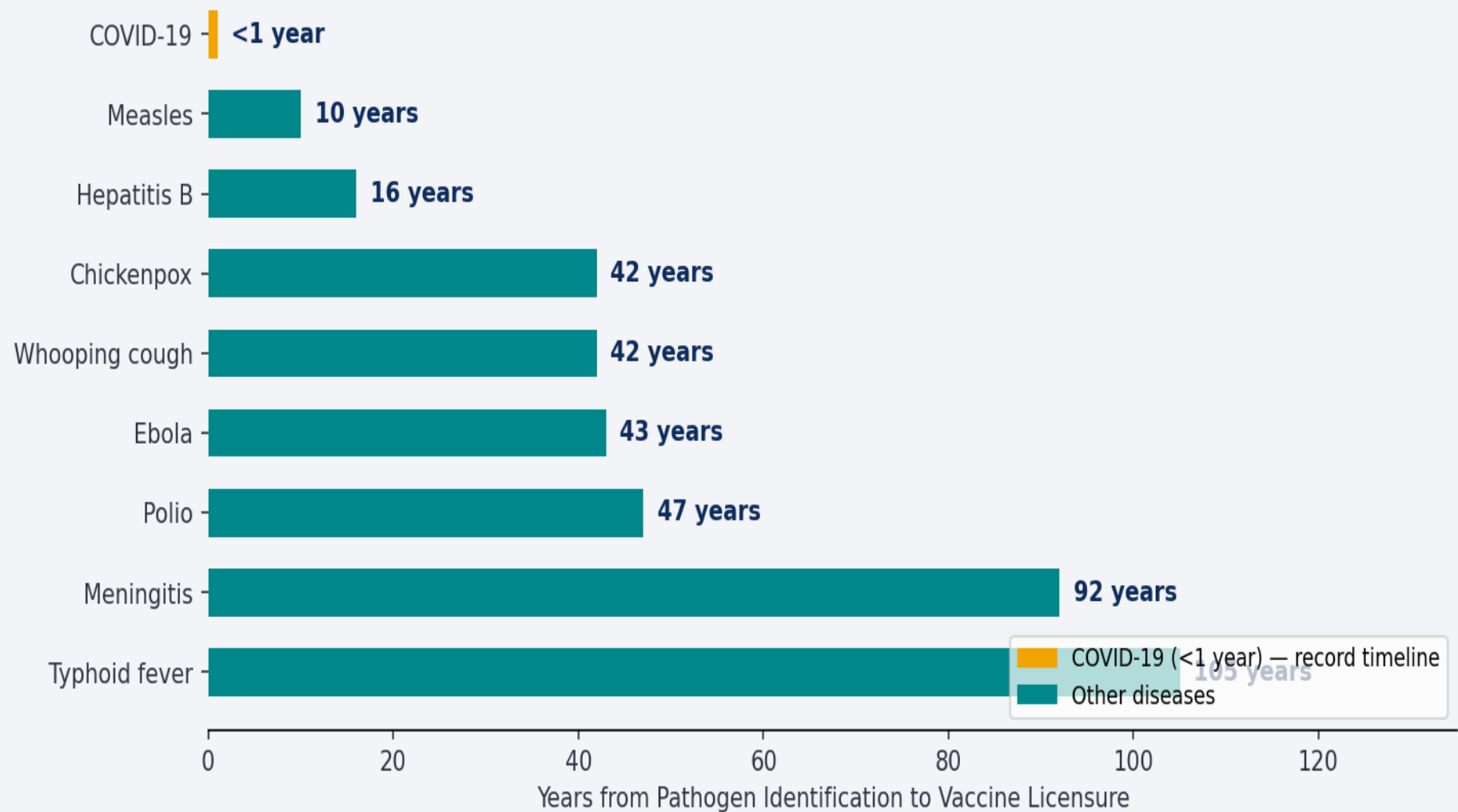
From Smallpox Eradication to COVID-19 in Record Time

Scope: 1796 – 2020 | A two-century journey of scientific innovation and global health impact

Data source: Our World in Data – ourworldindata.org/vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure



mRNA technology, genome sequencing & advanced microscopy collapsed development timelines from over a century to under a year.

COVID-19
<1 year
vs. Typhoid: 105 years

Source: Roush & Murphy (2007) via Our World in Data

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre-vaccine vs. current annual cases and deaths per million (US)

Disease	Pre-Vax Cases (per M/yr)	Post-Vax Cases (per M/yr)	Case Reduction	Pre-Vax Deaths (per M/yr)	Death Reduction
Diphtheria	158	0	100% ✓	13.7	100% ✓
Smallpox	250	0	100% ✓	2.9	100% ✓
Acute Polio	141	0	100% ✓	10	100% ✓
Measles	3,044	0.2	99.99% ✓	2.5	100% ✓
Rubella	242	0.04	99.98% ✓	0.09	100% ✓
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute Hep. B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5% !	24	31.3% !

 ✓ = 100% elimination achieved

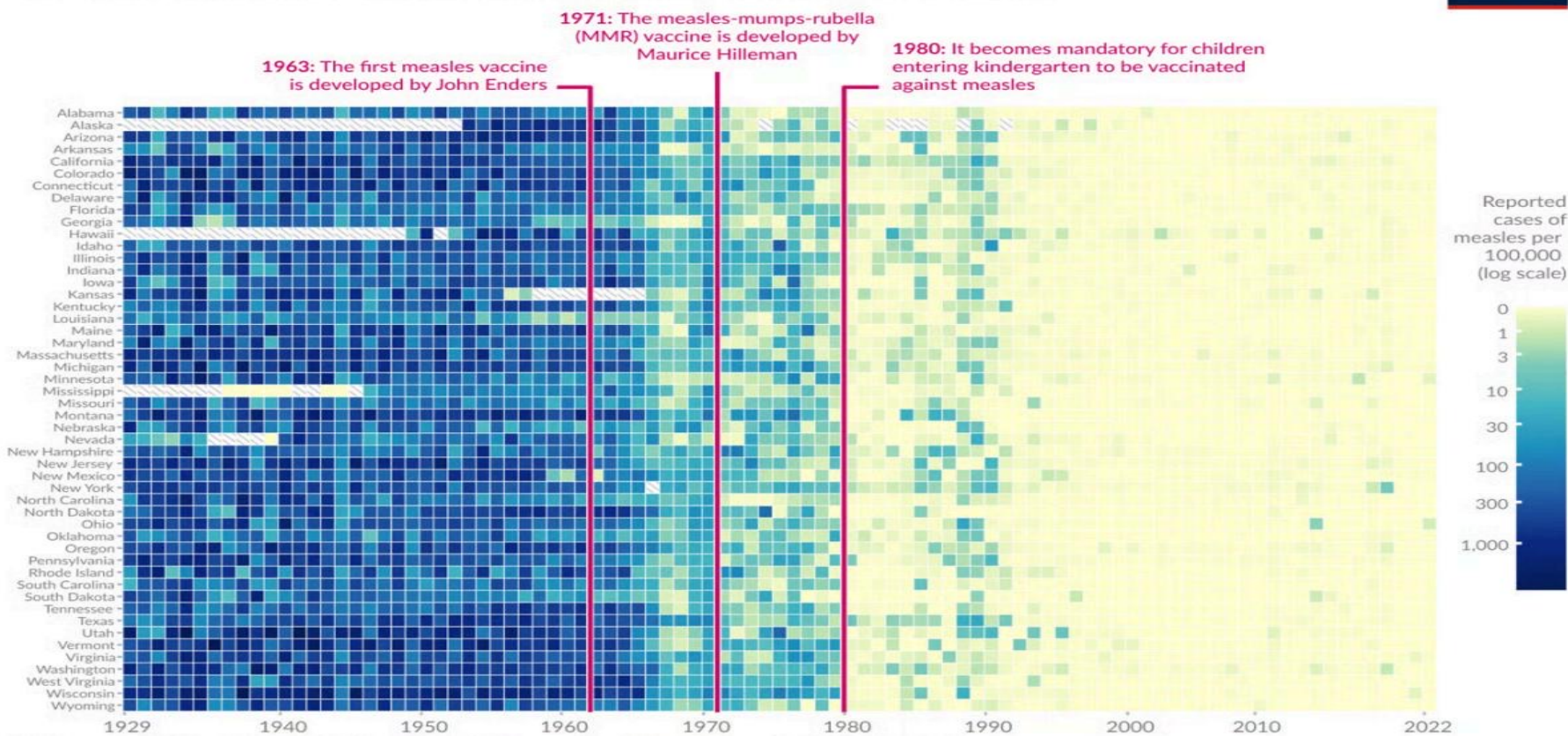
 ! = Incomplete control — remaining burden

Source: Roush & Murphy (2007) via Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Reported measles cases per 100,000 by state, 1929–2022

Vaccines reduced measles cases across US states



Data source: Project Tycho (2018); Centers for Disease Control and Prevention (1959–2022)

OurWorldinData.org — Research and data to make progress against the world's largest problems.

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1963

First measles vaccine (John Enders)

1971

MMR vaccine introduced (Maurice Hilleman)

1980

Kindergarten vaccination mandate — near-zero cases

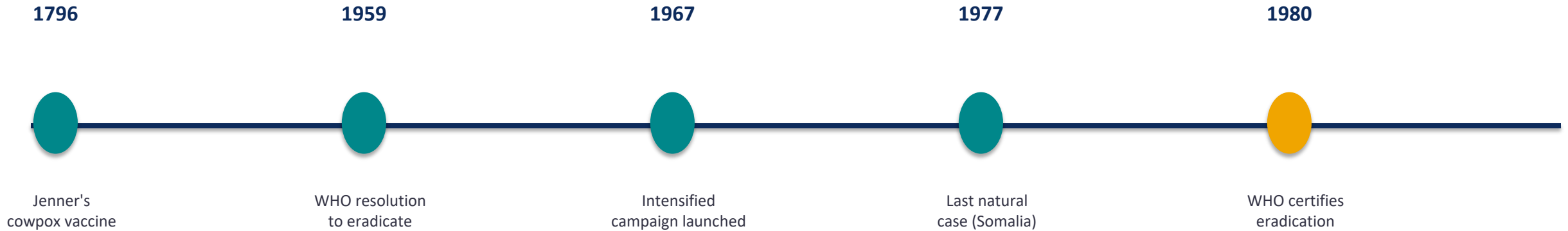
Before 1963: 100–1,000+ cases/100,000 in nearly every state.
After 1980: near-zero across all states.

Download image or data

Smallpox: The Only Human Disease Eradicated Through Vaccination

A 21-year global campaign — WHO resolution (1959) → last case (1977) → certification (1980)

Smallpox: 250 pre-vaccine cases/M/yr → 0 | 100% case & death reduction



Why Smallpox Was Uniquely Suited for Eradication:

- Infected humans only — no animal reservoir
- Visible, distinguishable symptoms enabled ring vaccination
- Cheap, stable, and effective vaccine

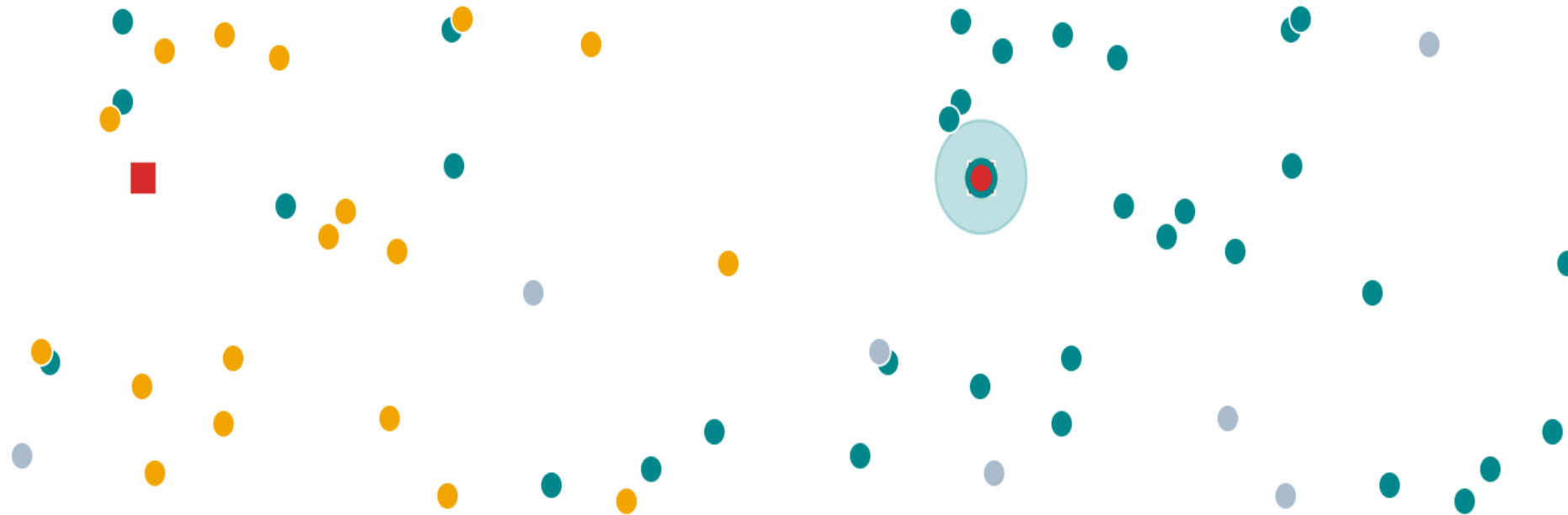
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Population-level vaccination shields those who cannot be vaccinated

Low Vaccination Coverage
(pathogen spreads freely)



High Vaccination Coverage
(herd immunity achieved)



Vaccines protect:

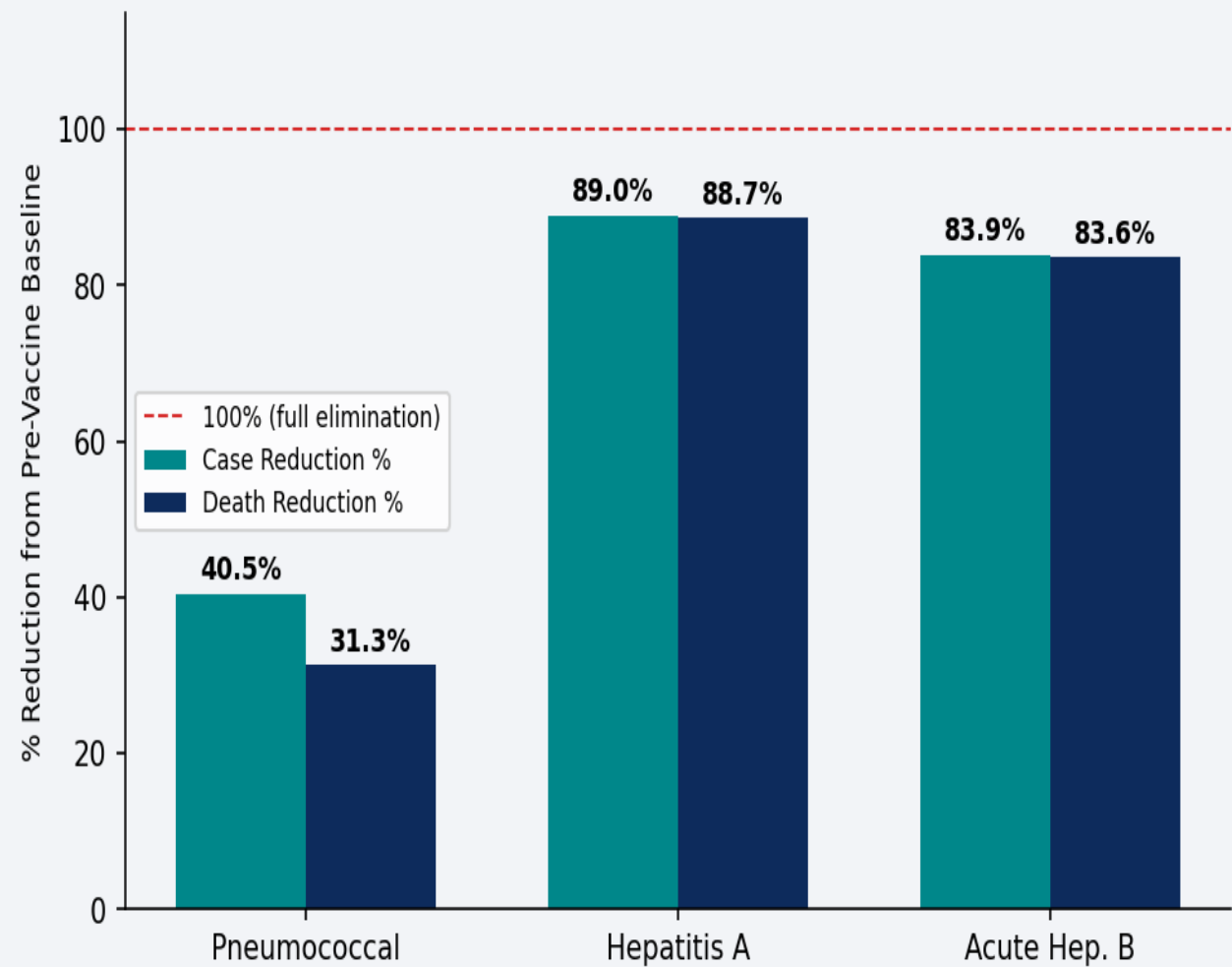
- Newborns
- Immunocompromised patients
- Cancer patients on chemotherapy

who cannot receive vaccines themselves.

● Vaccinated individual ■ Vulnerable (cannot be vaccinated) ● Infected / spreading disease ● Unvaccinated (susceptible)

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Not all vaccine-preventable diseases are under full control



Access Gaps

Not everyone globally has access to vaccines.

Supply Shortages

Shortages put children at risk in low-income regions.

Trust Deficits

Poor scientific understanding reduces uptake.

Diseases like pneumococcal, Hep A & B demonstrate that new vaccines and greater coverage are still urgently needed.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1 100% Elimination Achieved**
Diphtheria, smallpox, acute polio — zero cases and deaths in the US.
- 2 Development Timelines Collapsed**
From 105 years (typhoid) to under 1 year (COVID-19) via mRNA & genomic tools.
- 3 Measles Near-Zero Nationwide**
Across all 50 US states after the 1980 kindergarten vaccination mandate.
- 4 Herd Immunity Saves Lives**
High coverage protects newborns, immunocompromised, and chemo patients.
- 5 Coverage Gaps Remain**
Pneumococcal (40.5% case reduction), Hep A (89%), Hep B (83.9%) — work continues.

Call to Action for Global Health Systems

Expand vaccine access in low-income regions | Address supply chain shortages | Invest in science communication to rebuild public trust | Accelerate R&D for remaining high-burden diseases